

Regulated Document Decisioning

Governance & Operating-Model Packet

System: Regulated Document Decisioning (RDD) **Owner:** Christopher Williams **Platform:** UiPath Studio Web Maestro (BPMN agentic process) **Date:** May 14, 2026 **Version:** 1.0
Status: Operational end-to-end; hardening and observability roadmapped (see §6)

Purpose of this document

This packet is the model-risk and governance documentation for the RDD system. It is written to the standard a bank's model-risk function and its examiners would expect, and it is organized into four parts:

1. **Model Card** - what RDD is, what it decides, and its declared limitations
2. **Decision Logging Specification** - what is recorded for every decision and how
3. **Audit Retention Matrix** - what is retained, for how long, and under which regulatory driver
4. **Human Override Audit Trail Standard** - how human intervention is captured and scrutinized

A companion document, the *Governance & Operating-Model Memo*, provides the strategic and architectural overview. This packet is the controls-level detail beneath it.

A note on scope: RDD was built as a demonstration system on fictional institutions (Meridian Bank and its subsidiary Cascade Health). Where a control is implemented it is described as implemented; where it is designed but roadmapped it is marked as such. Nothing in this packet overstates the system's current state.

Part 1 - Model Card

1.1 System identity and intended use

RDD is a composite decisioning system. It ingests a loan application or an insurance claim submitted by email, renders a disposition, verifies that disposition against applicable regulatory frameworks, routes adverse or ambiguous outcomes through human review, and produces a statutory customer communication archived to an immutable audit store.

Intended use: automated first-pass decisioning and compliance verification of routine regulated documents, with mandatory human review for adverse, large, ambiguous, or data-flagged cases.

Out of scope / not intended use: RDD does not book loans or pay claims - it decisions and documents them. It is not a system of record for customer or policy data. It does not make a final adverse determination without either a deterministic policy basis or a human reviewer in the loop.

1.2 Composition - the "model" is a pipeline

RDD is not a single model. It is a five-stage pipeline of deterministic and LLM-driven components, and the distinction matters for model risk: the deterministic stages are auditable by inspection, while the LLM stages require behavioral governance. The five stages are:

1. **Document Classification Agent** (LLM). Classifies the submission as a loan application, an insurance claim, or unknown; confidence-scored, with a 0.70 floor below which it returns "unknown".
2. **Loan or Claim Intake** (deterministic script). Validates required fields, computes derived metrics (ratios, tiers, timeliness), and flags data-quality issues.
3. **Risk decisioning**, with two branch-specific implementations. For loans, **Credit Decisioning** (deterministic script) runs a rule ladder over credit score, loan-to-income, payment-to-income, and amount. For claims, the **Claims Adjudication Agent** (LLM) applies medical-necessity and policy-based reasoning, emits APPROVE, DENY, or REVIEW, and has an independent second-opinion tool available.
4. **Regulatory Compliance Agent** (LLM). Verifies the disposition against all applicable frameworks and produces evidence-citing findings.
5. **Customer Disclosure Agent** (LLM). Drafts the cover email and the statutory letter to regulatory specificity.

A human-review gate sits between stages 3 and 4. Deterministic merge scripts assemble the decision record between every stage.

1.3 Inputs and outputs

Inputs: a JSON payload delivered in an email body. Loan payloads carry applicant identity, requested amount, income, credit score, and credit-score age. Claim payloads carry member identity, policy number, claim amount, incident date, coverage and claim type, provider, and a claim description. Both now carry a customer email and a customer name.

Outputs: (a) a complete decisionEvent record - see Part 2; (b) a PDF statutory letter; (c) a customer email with the letter attached. The letter PDF is archived to an internal Storage Bucket.

1.4 Decision logic - deterministic stages

The **loan credit-decisioning ladder** evaluates in priority order; the first match wins:

1. Credit score below 580 → auto-decline (Reg B documented underwriting standard)
2. Loan-to-income ratio above 2.0 → auto-decline
3. Intake flagged for data-quality issues → manual review

4. Requested amount above \$50,000 → escalate for senior review
5. Loan-to-income ratio above 1.0 → escalate for senior review
6. Payment-to-income ratio above 0.40 → escalate for senior review
7. Credit score ≥ 720 , amount $\leq \$10,000$, payment-to-income ≤ 0.15 , loan-to-income ≤ 0.30 , credit data fresh → auto-approve
8. Otherwise → manual review

The ordering is intentional: hard knockouts (1–2) fire before amount-based escalation, so an unaffordable large request is declined rather than consuming a senior reviewer. This logic descends directly from the Week 1 loan-intake build.

1.5 Decision logic - LLM stages

The **Claims Adjudication Agent** adjudicates under five hard rules - intake-quality deference, medical-necessity reasoning, policy-based decisioning, escalation thresholds, and a no-invention discipline - and emits APPROVE, DENY, or REVIEW. It escalates any REVIEW above \$10,000 and any claim above \$50,000 regardless of disposition. It may consult an independent second-opinion tool on ambiguous or high-cost claims.

The **Regulatory Compliance Agent** does not decide - it verifies. It names the governing frameworks, produces a discrete finding against each, and sets a compliance status of compliant, requires-remediation, or non-compliant.

The **Customer Disclosure Agent** drafts communications under thirteen operating rules governing tone, statutory specificity, channel, and a no-invention discipline.

1.6 Regulatory frameworks the system operates under

- **Loan branch:** Reg B / ECOA (12 CFR § 1002), the Fair Credit Reporting Act, the Truth in Lending Act, and OFAC sanctions screening.
- **Claim branch:** ERISA Section 503 (29 U.S.C. § 1133), applicable state insurance codes, HIPAA, and OFAC sanctions screening.
- **Both branches:** SR 11-7 model risk management, which governs the decision record, human-override scrutiny, and this packet.

1.7 Declared limitations

- **LLM non-determinism on borderline cases.** The Claims Adjudication Agent can return different dispositions on identical borderline inputs across runs. This is mitigated by the human-review gate (borderline cases escalate) and is roadmapped for further mitigation by moving escalation determination into a deterministic step.
- **Sanctions screening is not live.** OFAC screening is currently a deferred/mock control; the compliance agent records its status honestly as deferred rather than asserting a clear result. A live watchlist service is a Phase 2 item.
- **Decision-vocabulary divergence.** The loan and claim branches currently emit different disposition enums; downstream consumers handle both. Unification is roadmapped.

- **Retention not yet configured.** The retention policy in Part 3 is the designed standard; the Storage Bucket retention setting is roadmapped (see §6).
 - **Single-tenant, demo-scale.** RDD runs in a single solution against fictional institutions. A production deployment would partition by line of business and integrate real systems of record.
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Part 2 - Decision Logging Specification

2.1 Principle

Every RDD run produces exactly one decision record - the `decisionEvent` object - under one stable identifier. The record is **append-only by construction**: each pipeline stage spreads the existing record and extends it with its own output. No stage destroys or rewrites a prior stage's contribution. The record that reaches the end of the pipeline contains the full, ordered history of the decision.

2.2 The decision identifier

The identifier is minted once, at intake, in the form `RDD-<YYYYMMDD>-<six base36 characters>` - for example `RDD-20260514-V1SN7X`. It is date-stamped, sortable, human-readable, and unique. It flows unchanged through every stage and becomes the `reference_number` printed on the customer's letter and cited in the customer email. One decision, one identifier, appearing identically in the internal record and every customer-facing artifact.

2.3 Record structure

The `decisionEvent` record carries the following top-level fields, each tagged with the pipeline stage that writes it:

- **decision_event_id** (Intake) - the stable identifier from §2.2.
- **created_at** (Intake) - record creation timestamp. *Field present; population roadmapped.*
- **event_type** (Classification) - one of `loan_application`, `insurance_claim`, or `unknown`.
- **source_channel** (Intake) - ingestion channel. *Field present; population roadmapped.*
- **classification_output** (Classification) - event type, confidence score, classifier reasoning, and timestamp.
- **intake_output** (Intake) - structured payload, derived metrics, data-quality issues, and intake status.
- **risk_output** (Risk decisioning) - disposition, reasoning, thresholds applied, sanctions status, policy version, and timestamp.
- **compliance_output** (Compliance) - compliance status, frameworks applied, evidence-citing findings, adverse-action flag, required disclosures, and timestamp.

- **communication_output** (Disclosure and archival) - drafted email and letter, channel, priority, drafted reasoning, plus archival metadata (bucket path, filename, document ID, archive timestamp).
- **final_disposition** (Human-review merge or auto-disposition step) - one of approved, declined, or unknown.
- **human_touchpoints** (Human-review merge) - array of human intervention records; see Part 4.
- **reference_number** (Disclosure merge) - equals decision_event_id; the customer-facing reference.

2.4 What each stage logs

- **Classification** records the event type, the confidence score, and the classifier's reasoning - so a misroute can be reconstructed and audited.
- **Intake** records the full structured payload, every derived metric, and an explicit list of data-quality issues. A flagged intake is itself an auditable event.
- **Risk decisioning** records the disposition, the specific reasons, the thresholds applied, the policy version in force, and - for loans - the exact rule that fired.
- **Compliance** records a discrete finding per applicable framework. Every finding must cite specific evidence from the case; generic assurances are explicitly disallowed by the agent's operating rules.
- **Disclosure** records both customer-facing artifacts verbatim, plus the agent's internal drafting rationale (retained for audit, never shown to the customer).

2.5 Storage and immutability

Today the decision record lives in process state and process logs, and the letter PDF is archived to the rdd-decision-letters-archive Storage Bucket under the content-addressed path adverse-actions/<reference_number>.pdf. Phase 2 promotes the decision record to a first-class UiPath Data Fabric entity, making it queryable and giving the audit trail a durable, structured home independent of process logs. Tamper-evidence via content hashing of the archived PDF is roadmapped.

Part 3 - Audit Retention Matrix

3.1 Retention envelope

RDD applies a single conservative retention envelope of **six years** across all decision artifacts. This is the longest applicable obligation: it covers ERISA's record-retention horizon for benefit-claim records and comfortably exceeds Reg B's 25-month requirement for consumer-credit adverse-action records. Applying the longest envelope uniformly avoids per-artifact retention logic and guarantees the audit trail outlives any examination window.

3.2 The artifacts

Each of the six artifacts below sits under the six-year envelope established in §3.1. The driver, storage location, and current implementation status follow.

Decision record (decisionEvent). Driven by SR 11-7, Reg B, and ERISA § 503. Lives in process logs today; promoted to a UiPath Data Fabric entity in Phase 2. *Operational; durable store roadmapped.*

Adverse-action letter (PDF). Driven by Reg B 12 CFR § 1002.12 (25-month minimum) and ERISA § 503. Archived to the rdd-decision-letters-archive Storage Bucket. *Archival operational; storage-layer retention setting roadmapped.*

Customer email record. Retained as supporting evidence of delivery. Stored in the mail system and archive. *Operational.*

Human-review task and reviewer rationale. Driven by SR 11-7. Stored in Action Center and on decisionEvent.human_touchpoints. *Operational.*

Compliance findings. Examiner-facing record. Stored on decisionEvent.compliance_output. *Operational.*

Classification and intake records. Support reconstruction of routing and data quality. Stored on the decisionEvent. *Operational.*

3.3 Outstanding action

The Storage Bucket retention policy must be set to six years in Orchestrator. Until then, archival is operational but retention duration is not enforced at the storage layer. This is the single highest-priority item in the hardening backlog from a governance standpoint.

Part 4 - Human Override Audit Trail Standard

4.1 Principle (SR 11-7)

Human judgment is preserved in RDD, and it is audited. Every human intervention is recorded, the reviewer's rationale is required, and any override of the system's recommendation is flagged and independently scrutinized by the Regulatory Compliance Agent. A human can change any decision - but never silently.

4.2 When a human touchpoint occurs

The human-review gate fires when the disposition is a review or escalation value (manual_review, escalate_senior_review, or REVIEW) or when the escalation flag is set. The case suspends and is assigned to a reviewer through a dedicated Action Center application; the process instance remains alive - for days if necessary - until the reviewer responds.

4.3 What is captured

Each intervention appends a record to `human_touchpoints` containing the following fields:

- **reviewer_action** - the reviewer's decision, either approve or reject.
- **reviewer_comment** - the reviewer's free-text rationale.
- **reviewed_at** - timestamp of the decision.
- **agent_decision_at_review** - the disposition the system had produced before the human acted.
- **was_override** - true if the human reversed the system's recommendation.
- **agent_abstained** - true if the system declined to make a call (a review or escalation disposition) and the human made the determination.

4.4 Override vs. abstention

The standard distinguishes two materially different events:

- **Override** - the system produced a clear recommendation (approve or decline) and the reviewer reversed it. This is the highest-scrutiny event: a human contradicting the model.
- **Abstention** - the system produced a review/escalation disposition, i.e. it explicitly declined to decide, and the reviewer made the determination. This is the system working as designed.

`was_override` is set true only on a genuine reversal - system-approve reversed to reject, or system-decline reversed to approve. `agent_abstained` captures the abstention case. The two flags make every human touchpoint self-classifying in the audit record.

4.5 Independent scrutiny

The Regulatory Compliance Agent inspects every human touchpoint:

- On an **override**, it raises an SR 11-7 finding requiring that the reviewer's rationale be retained for model-risk audit, and it quotes the reviewer's comment as evidence. If the reviewer's comment is empty, it marks the case requires-remediation - an override without a recorded rationale is itself a control gap.
- On an **abstention**, it raises a finding noting the agent abstained and the reviewer determined the outcome, citing the reviewer's comment as evidence.

This closes the loop: the human decision is not only recorded, it is checked, by a separate component, against a documented standard.

4.6 Chain of custody

For any decision, the full custody chain is reconstructable from the single decision record: the system's recommendation (`risk_output`), whether and how a human intervened (`human_touchpoints`), whether that intervention was an override or an abstention (the two flags), the reviewer's rationale (`reviewer_comment`), the compliance assessment of

that intervention (compliance_output), and the final outcome communicated to the customer (communication_output and final_disposition).

§5 - Governance posture summary

RDD's governance posture rests on five implemented controls and one principle.

Implemented controls: (1) a single immutable decision record per decision; (2) a stable identifier flowing from intake to customer letter; (3) mandatory human review for adverse, large, ambiguous, and data-flagged cases; (4) explicit, evidence-citing compliance verification against named frameworks; (5) independent scrutiny of every human override.

The principle: governance is an *output* of the pipeline, not a checkpoint within it. Every decision RDD makes produces, as a byproduct of running, the complete record an examiner would ask for.

§6 - Hardening and roadmap

Near-term hardening (non-blocking): configure Storage Bucket retention to six years (highest governance priority); move escalation determination into a fully deterministic step; unify the loan and claim disposition vocabularies; content-hash archived PDFs for tamper-evidence; populate created_at and source_channel; resolve templated placeholders against a branch-context lookup.

Phase 2 - observability: promote the decision record to a UiPath Data Fabric entity; build an operations dashboard for compliance and operations staff to query decisions by type, disposition, and framework, and drill into any archived letter; replace the mocked sanctions control with a live watchlist service; add a regulatory-citation service.

Phase 3 - completion of this packet: this document is the first issue. Subsequent issues will add per-component performance characteristics once production telemetry exists, and a formal change-control log for policy thresholds and agent prompts.

End of packet - Version 1.0, May 14, 2026.